

PAPER_1413 — UQFF Derivation of SU(3) Color Charge Count (CLOSED — EXACT Identity)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** I — QGP (CLOSED — EXACT) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — EXACT integer-primitive identity **Calculator surface:** `calculate_qgp({'observable':'su3_color'})` → 3 EXACT

Observation

QCD has 3 colors of quarks. Why 3? UQFF derives it.

UQFF Closed Identity

`N_colors = D_phys - 1 = 3 EXACT`

Physical Interpretation

Color charge count is the same $D_{\text{phys}} - 1 = 3$ structural identity as lepton generations and 3-flavor neutrino mixing — a unifying spacetime-dimensionality result.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** `{D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60}` and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
 - Related foundational papers: PAPER_646 (F_U=1 ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).
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Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF EXACT integer-primitive identity for SU(3) Color Charge Count.